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Information processing apparatus that executes control for enabling a security setting related to encrypted communication when a function related to a push scan is enabled, image processing apparatus, information processing method, and storage medium

Abstract

There is provided with an information processing apparatus. A controlling unit, in a case where a condition related to at least any of a mode of communication with an image processing apparatus, a form of a connection with an image processing apparatus, and an authorization level of communication with an image processing apparatus is satisfied, enables an instruction for a push scan to the image processing apparatus. A sending unit transmits to the image processing apparatus a credential to be used in transmission processing in a push scan. The controlling unit, in a case where a condition related to at least any of the mode of communication, the connection form, and the authorization level is not satisfied, controls to not perform an instruction for a push scan to the image processing apparatus.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/687,954, filed Mar. 7, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an information processing apparatus, an image processing apparatus, an information processing method, and a storage medium.

Description of the Related Art

(2) In recent years, a configuration in which a push scan request is transmitted from a client terminal to a scanner terminal and scanned data is transmitted from the scanner terminal to an external terminal has begun to become widespread (Japanese Patent Laid-Open No. 2017-112508). In such a system, first, a user sets an original in a scanner terminal, specifies a destination, a storage location, a scan resolution, and other settings for storing scan results from a client terminal, and selects to start a scan. Designated information and a scan start instruction are transmitted from the client terminal to the scanner terminal, and the scanner terminal, after having received the information, performs the scan. After that, the scanner terminal connects to a designated destination terminal and transmits the scanned data.

(3) Although various methods have been proposed for such scanning protocols, HTTP-based IPP Scan (PWG 5100.17) and eSCL protocols have become popular. In addition, a search protocol such as mDNS (RFC 6762) is commonly used as a protocol for retrieving and registering a scanner terminal in a client terminal.

SUMMARY OF THE INVENTION

(4) According to one embodiment of the invention, an information processing apparatus which executes an application which uses a predetermined protocol that supports a scanning method for both an instruction for a pull scan and an instruction for a push scan over a network, the apparatus comprises: a controlling unit configured to, in a case where a condition related to at least any of a mode of communication with an image processing apparatus, a form of a connection with an image processing apparatus, and an authorization level of communication with an image processing apparatus is satisfied, enable an instruction for a push scan to the image processing apparatus; and a sending unit configured to transmit to the image processing apparatus a credential to be used in transmission processing in a push scan, wherein the controlling unit, in a case where a condition related to at least any of the mode of communication, the connection form, and the authorization level is not satisfied, controls to not perform an instruction for a push scan to the image processing apparatus.

(5) According to another embodiment of the invention, an image processing apparatus which accepts an instruction from an information processing apparatus by an application which uses a predetermined protocol that supports a scanning method for both an instruction for a pull scan and an instruction for a push scan over a network, the image processing apparatus comprises: a determination unit configured to determine whether or not communication with the information processing apparatus is encrypted; and a sending unit configured to transmit information indicating whether or not a push scan is possible in the information processing apparatus in accordance with whether or not the communication is encrypted.

(6) According to still another embodiment of the invention, an image processing apparatus which accepts an instruction from an information processing apparatus by an application which uses a predetermined protocol that supports a scanning method for both an instruction for a pull scan and an instruction for a push scan over a network, the image processing apparatus comprises: a first

setting unit configured to set whether or not to encrypt communication with the information processing apparatus; and a second setting unit configured to set whether or not to enable a push scan in accordance with the setting as to whether or not to encrypt the communication.
(7) According to yet another embodiment of the invention, an information processing method performed by an information processing apparatus which executes an application which uses a predetermined protocol that supports a scanning method for both an instruction for a pull scan and an instruction for a push scan over a network, the information processing method comprises: enabling, in a case where a condition related to at least any of a mode of communication with an image processing apparatus, a form of a connection with an image processing apparatus, and an authorization level of communication with an image processing apparatus is satisfied, an instruction for a push scan to the image processing apparatus; and transmitting to the image processing apparatus a credential to be used in transmission processing in a push scan, wherein the enabling, in a case where a condition related to at least any of the mode of communication, the connection form, and the authorization level is not satisfied, controls to not perform an instruction for a push scan to the image processing apparatus.
(8) According to still yet another embodiment of the invention, an information processing method performed by an image processing apparatus which accepts an instruction from an information processing apparatus by an application which uses a predetermined protocol that supports a scanning method for both an instruction for a pull scan and an instruction for a push scan over a network, the information processing method comprises: determining whether or not communication with the information processing apparatus is encrypted; and transmitting information indicating whether or not a push scan is possible in the information processing apparatus in accordance with whether or not the communication is encrypted.
(9) According to yet still embodiment of the invention, an information processing method performed by an image processing apparatus which accepts an instruction from an information processing apparatus by an application which uses a predetermined protocol that supports a scanning method for both an instruction for a pull scan and an instruction for a push scan over a network, the information processing method comprises: setting whether or not to encrypt communication with the information processing apparatus; and setting whether or not to enable a push scan in accordance with the setting as to whether or not to encrypt the communication.
(10) According to still yet another embodiment of the invention, a non-transitory computer-readable storage medium stores a program which, when executed by a computer comprising a processor and a memory, executes an application which uses a predetermined protocol that supports a scanning method for both an instruction for a pull scan and an instruction for a push scan over a network, and causes the computer to: enable, in a case where a condition related to at least any of a mode of communication with an image processing apparatus, a form of a connection with an image processing apparatus, and an authorization level of communication with an image processing apparatus is satisfied, an instruction for a push scan to the image processing apparatus; and transmit to the image processing apparatus a credential to be used in transmission processing in a push scan, wherein control, in a case where a condition related to at least any of the mode of communication, the connection form, and the authorization level is not satisfied, to not perform an instruction for a push scan to the image processing apparatus.
(11) Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram illustrating an exemplary system including a processing terminal according

to a first embodiment of the present invention.
(2) FIGS. **2**A through **2**D are diagrams illustrating examples of packets transmitted/received by the processing terminal according to the first embodiment.
(3) FIGS. **3**A through **3**F are diagrams illustrating examples of requests transmitted by the processing terminal according to the first embodiment and responses thereto.
(4) FIG. **4** is a flowchart illustrating an example of a search process in the system according to the first embodiment.
(5) FIG. **5** is a flowchart illustrating an example of a scan process in the system according to the first embodiment.
(6) FIG. **6** is a block diagram illustrating an example of a hardware configuration of the processing terminal according to the first embodiment.
(7) FIG. **7** is a diagram illustrating an example of a flow of processing terminal screens and confirmation screens according to the first embodiment.
(8) FIG. **8** is a flowchart illustrating an example of a scan process that the processing terminal performs according to the first embodiment.
(9) FIG. **9** is a flowchart illustrating an example of a confirmation process that the processing terminal performs according to the first embodiment.
(10) FIG. **10** is a view illustrating an example of a flow of registration screens that the processing terminal according to a second embodiment displays.
(11) FIG. **11** is a flowchart illustrating an example of a search process that the processing terminal performs according to the second embodiment.
(12) FIG. **12** is a flowchart illustrating an example of a registration process that the processing terminal performs according to the second embodiment.
(13) FIG. **13** is a flowchart illustrating an example of a scan process that the processing terminal performs according to a third embodiment.
(14) FIG. **14** is a view illustrating an example of a screen that the image processing apparatus according to the fourth embodiment displays.
(15) FIG. **15** is a flowchart illustrating an example of a setting process that the image processing apparatus performs according to the fourth embodiment.
(16) FIG. **16** is a block diagram illustrating an example of a hardware configuration of an image processing apparatus according to the fourth embodiment.
(17) FIGS. **17**A through **17**C are diagrams illustrating examples of requests transmitted by the processing terminal according to a fifth embodiment and responses thereto.

DESCRIPTION OF THE EMBODIMENTS

(18) As destinations for storing scan data, various external terminals such as servers in the same LAN, the self terminal which has instructed the scan, and storage of a cloud service can be specified. Authentication is required for the scanner terminal to connect to these external terminals, and the authentication information is also transmitted from the client terminal to the scanner terminal together with the scan start instruction request. Authentication information is information such as a token, a user name, and a password, for example. However, if the communication path to the scanner terminal from the client terminal is not encrypted, there is a risk that such authentication information will be eavesdropped. When the authentication information is eavesdropped, there is a problem that the user may be spoofed, the external terminal accessed, and confidential data stored in the storage extracted and leaked.
(19) One embodiment of the present invention prevents authentication information from being eavesdropped when the processing terminal transmits a scan instruction including authentication information.
(20) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made an invention that

requires all such features, and multiple such features may be combined as appropriate.
(21) Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.
First Embodiment
(22) FIG. **1** is a diagram illustrating an example of a configuration of a printing system including a processing terminal **101** which is an information processing apparatus according to the present embodiment. The printing system includes a processing terminal **101**, image processing apparatuses **102** to **104** equipped with a scanning function, and a cloud storage service (the service) **105**. The processing terminal **101** communicates via a network **100** with the image processing apparatuses **102** to **104** that reside in the same LAN. The network **100** can transmit and receive data between the processing terminal **101** and the image processing apparatuses **102** to **104**, and any physical layer communication method may be adopted. The image processing apparatuses **102** to **104** provide a scanning function and communicate with the service **105** on the Internet via a communication network or cellular network. Hereinafter, when simply referred to as an image processing apparatus, any one of the image processing apparatuses **102** to **104** is used.
(23) The network **100** may be, for example, a communication network such as a LAN or WAN, a cellular network (e.g., LTE or 5G), or a wireless network that is compliant with IEEE 802.11, and may be configured by combining these communications. The processing terminal **101** may be any terminal, such as a desktop personal computer, a tablet, or a mobile phone terminal, that can be operated by acquiring input from a user. The image processing apparatus **102** is not particularly limited as long as it has a scan function, and may be, for example, a device of a single scanner or a multifunction peripheral having a print function.
(24) The processing terminal **101** according to the present embodiment transmits a push scan start request to the image processing apparatus through the network **100** and executes a scan. Upon receiving the push scan start request, the image processing apparatus uses the authentication information included in the packet of the push scan start request to connect to the service **105** which is the designated external destination, and transmits the scanned data.
(25) [Push Scan]
(26) Next, transmission and reception of communication for a typical push scan will be described with reference to FIGS. **2** to **5**. In the present embodiment, the push scan instruction described below is performed by communication in an HTTP protocol using XML, and a scanner terminal search is performed by the communication in an mDNS protocol. FIGS. **2**A to **2**D are views illustrating examples of requests and responses transmitted/received in the mDNS protocol.
(27) FIG. **4** is a diagram illustrating an example of a sequence in which a user searches for an image processing apparatus by using the processing terminal **101** and registers the image processing apparatus in the processing terminal **101**. In step S**401**, the user selects the “search” button from the operation screen of the processing terminal **101** to initiate the subsequent processing continuing from step S**402**. In step S**402**, the processing terminal **101** transmits a request (a search request) as illustrated in FIG. **2**A to determine whether or not a terminal in which the scan service is enabled exists in the same link network using mDNS protocol as a multicast packet.
(28) In the processing according to the present embodiment, a plaintext scan service protocol uses port number **80**, and a TLS encrypted scan service protocol uses port number **443**. When both plaintext communication and encrypted communication are enabled in communication with the processing terminal **101**, the image processing apparatus, after receiving the search request, performs an mDNS response (search response) including both port number **80** and port number **443** as illustrated in FIG. **2**B. In addition, as illustrated in FIG. **2**C and FIG. **2**D, the image processing apparatus, in which only one of them is enabled, performs a search response including the service of only the enabled one of port number **80** and port number **443**. TLS is communication using TCP/IP, and if a setting for TLS is enabled, communication between the processing terminal **101**

and the image processing apparatus **102** is encrypted; if it is disabled, plaintext communication is performed.

(29) Step S**403** to step S**404** is a process in which the image processing apparatus that has received the search request from the processing terminal **101** returns a response. In this example, the image processing apparatus in step S**403** transmits an mDNS response as illustrated in FIG. **2**C to the processing terminal **101**, and the image processing apparatus in step S**404** transmits an mDNS response as illustrated in FIG. **2**B or FIG. **2**D. Following step S**403** and step S**404**, the process proceeds to step S**405**.

(30) In step S**405**, the processing terminal **101**, after having received the search response, displays a list of image processing apparatuses that returned a search response on the display unit. In step S**406**, the processing terminal **101** acquires a desired selection from the list of image processing apparatuses by user input.

(31) In step S**407**, the processing terminal **101** may transmit a request (detail request) for obtaining more detailed information to the image processing apparatus selected in step S**406** in order to know what kind of scanning is possible. In step S**408**, the processing terminal **101** receives a response to the detailed request from the image processing apparatus. In step S**409**, the processing terminal **101** performs a process of registering the selected image processing apparatus in internal memory, stores information indicating the selected image processing apparatus in the storage area, and terminates the registration process. In the present embodiment, the information indicating the image processing apparatus for which the registration process is completed is stored in a volatile region of the processing terminal **101**. This storage state is maintained even when the power of the processing terminal **101** is turned off, and can be referred to and operated by the user at an arbitrary timing.

(32) FIG. **5** is a diagram illustrating an example of a sequence in which the processing terminal **101** performs a push scan instruction based on a user operation. In this example, the processing illustrated in FIG. **5** is performed after the image processing apparatus **102**, in which only plaintext communication is enabled, is selected in step S**406**. In step S**501**, the user selects the scan start button from the operation screen of the processing terminal **101** with a scan image original set in advance in the image processing apparatus **102**, and the processing following the subsequent step S**402** is started.

(33) In response to the selection of the scan start button in step S**502**, the processing terminal **101** transmits a scan start request as illustrated in FIG. **3**A to the image processing apparatus **102**. In step S**503**, the image processing apparatus **102** transmits a response to the scan start request to the processing terminal **101**, as illustrated in FIG. **3**B. In the example of FIG. **3**A, the “DestinationURI” attribute indicates a destination of the push scan to which the scan data is to be stored by POST indicated in “HttpMethod”. The example of FIG. **3**A also illustrates the use of an attribute indicated by “JobPassword” as an authentication token when connecting to a destination. The destination of the (push) scan is a destination to which the scan data generated by the (push) scan is transmitted, and in the present embodiment, the service **105** is designated. The required authentication method differs in accordance with authentication settings of the destination, and, for example, OAuth authentication, DIGEST authentication, BASIC authentication, or the like are employed.

(34) In step S**504**, the image processing apparatus **102** scans the original in response to receiving the scan start request. In step S**505**, the image processing apparatus **102** transmits a connection request to the destination specified in the scan start request (here, the service **105**). Here, the image processing apparatus **102** adds the required authentication information and transmits a connection request as illustrated in FIG. **3**C. The image processing apparatus **102** according to the present embodiment transmits a connection request including authentication information of BASIC authentication, but any authentication information corresponding to an authentication method such as a token for OAuth authentication may be used. In step S**506**, the image processing apparatus **102**

receives a successful connection response from the service **105** for which the authentication was successful.
(35) In step S**507**, the image processing apparatus **102** transmits scan data obtained by scanning an original to the service **105**, and receives a response indicating completion of reception of the scan data in step S**508**. In addition, the processing terminal **101** periodically transmits a query request for the scan job status as illustrated in FIG. **3**E to the image processing apparatus **102**. Upon receiving the query request, the image processing apparatus **102** transmits a scan job status response as illustrated in FIG. **3**F to the processing terminal **101**, and when the job is stored normally in the destination, transmits a response indicating that the storage is completed to the processing terminal **101**.
(36) In this example, since only plaintext communication is enabled for the image processing apparatus **102**, the exchange illustrated in FIG. **3**A and FIG. **3**B is performed in plaintext on the HTTP port **80**. Therefore, since an authentication token or the like described above is transmitted as plaintext, there is a problem in that the packet may be eavesdropped and the authentication information may be leaked. Incidentally, eavesdropping is easy to prevent in the case of transmission and reception using encrypted communication of the HTTPS port **443**.
(37) Therefore, the processing terminal **101** determines whether or not a mode of communication with the image processing apparatus **102**, the connection condition, or an authorization level satisfies a predetermined condition, and performs control so as not to enable an instruction for a push scan to the image processing apparatus **102** if the condition is not satisfied. Here, when the predetermined condition regarding the communication mode is not satisfied and the communication with the image processing apparatus **102** is not encrypted, the processing terminal **101** according to the present embodiment restricts the start of the push scan by the image processing apparatus **102**. When communication with the image processing apparatus **102** is encrypted, the processing terminal **101** can transmit credentials to be transmitted to the service **105** in a push scan to the image processing apparatus **102** without restricting a push scan from being started. The credentials are authentication information required for user authentication such as BASIC authentication or DIGEST authentication, and are transmitted by the image processing apparatus **102** in the above-described step S**505**. The following description assumes that a user name and a password are used as the authentication information.
(38) Further, as a case where the predetermined condition related to the connection condition is satisfied, the processing terminal **101** does not restrict the start of the push scan when the communication path with the image processing apparatus **102** is a P2P connection of a wireless LAN such as with WiFi Direct. This is because, in the P2P connection of the wireless LAN, since other terminals cannot participate in the connection and the wireless LAN layer is used, there is less risk of eavesdropping even if the plaintext communication of HTTP is performed.
(39) FIG. **6** is a block diagram illustrating an example of the hardware configuration of the processing terminal **101**. The processing terminal **101** includes a CPU **601**, a ROM **602**, a RAM **603**, a storage unit **604**, an operation unit **605**, and a communication unit **606**. The CPU **601** directly or indirectly controls each device (such as the ROM and the RAM) connected by an internal device and executes a program for implementing the invention. The ROM **602** is a read-only storage device that stores programs executed by the CPU **601** and stores BIOS as firmware. The RAM **603** functions as the main memory or work memory of the CPU **601** and is utilized to load software modules for implementing the invention. The storage unit **604** is a storage area and is, for example, a hard disk drive (HDD) or a solid state drive (SSD) in which an OS or a software module that is basic software is stored. The operation unit **605** functions as a display unit for displaying information to the user and a reception unit for receiving an instruction from the user. The operation unit **605** is, for example, a liquid crystal display unit having a touch panel function or a display having various hard keys.
(40) The CPU **601** controls the display of information and the reception of user operations in

cooperation with the operation unit **605**. The communication unit **606** is an interface for the processing terminal **101** to connect to the network. The communication unit **606** according to the present embodiment is assumed to be a communication interface that performs wired communication based on Ethernet (registered trademark), but is not particularly limited thereto as long as communication is enabled. The communication unit **606** may be, for example, a wireless communication interface conforming to IEEE 802.11 series. The communication unit **606** may perform communication as a wireless communication interface. Further, for example, the communication unit **606** may perform communication by a 3G line such as CDMA, a 4G line such as LTE, or mobile communication such as 5G NR. Although the respective processes performed by the CPU **601** according to the present embodiment are described as being realized by the processing terminal **101** which is dedicated hardware, some or all of the processes may be performed by a separate computer.
(41) Next, with reference to FIGS. **7** to **9**, a control process by the processing terminal **101** according to the present embodiment for restricting the start of the push scan in an unencrypted plaintext communication mode will be described. FIG. **7** illustrates an example of a screen flow of the operation unit **605** displayed by the processing terminal **101** according to the present embodiment. Screen **701** is an example of a screen for displaying a list of image processing apparatuses registered in the processing terminal **101**. When the user selects any terminal that the user wishes to use to perform a scan from the image processing apparatuses displayed on the screen **701**, the screen transitions to the screen **702**. The screen **702** is a main menu screen for the selected image processing apparatus, and displays a state of the image processing apparatus (for example, an idle state or a busy state) or a button for performing detailed setting for scanning. In this example, when the user presses on the button labeled “Open Scan Setting”, a screen **703**, which is a screen for performing detailed settings, is displayed. In the example of FIG. **7**, the screen **703** is a screen for setting a size, a resolution, or a format for scanning, but any setting may be made as long as it is used for scanning such as a position of a start point of scanning or a feed direction of an original, for example.
(42) When the user selects an item on the screen **703**, the operation unit **605** displays a screen for performing detailed settings on the selected item. Screens **704** to **708** are examples of screens for settings corresponding to respective items selected on the screen **703**. Screen **704** is a screen for setting the destination of the push scan. In the screen **704**, it is possible to set whether the destination for storing the scan data is the self terminal or an external terminal, a path for storing the scan data, and authentication information necessary for connecting to the destination. The item “Destination” is displayed on the screen **704**, and the setting of whether the destination is the self terminal or an external terminal and the setting of the detail (URL) when the destination is an external terminal are inputted. The screen **704** displays, as the authentication information, a form for inputting the authentication information for a user authentication request requiring a user name and a password, such as BASIC authentication or DIGEST authentication from the destination terminal. The authentication information may be set in advance, and when the image processing apparatus **102** transmits a request for user authentication to the processing terminal **101**, a screen prompting input of the authentication information may be displayed as a pop-up on the operation screen of the processing terminal **101**. Further, when the user authentication for the service **105** based on the authentication information has already been completed, an item for setting whether or not the token stored in the processing terminal **101** is transmitted to the image processing apparatus **102** may be provided on the screen **703**. In this embodiment, various protocols such as HTTP, FTP, or SMB may be used as the protocol setting for connecting to the service **105**, and parameters required for setting the destination (here, the service **105**) may be optionally changeable.
(43) FIG. **8** is a flowchart illustrating an exemplary process performed by the CPU **601** of the processing terminal **101** according to the present embodiment to restrict the start of push scanning by displaying a warning screen. When the “Scan” button is pressed in the screen **703**, the CPU **601**

of the processing terminal **101** starts the processing of step S**801**, and advances the processing to step S**802**.
(44) In step S**802**, the CPU **601** determines whether scanning can be started. Here, the CPU **601** first determines whether the scanning process performed by the image processing apparatus **102** is a push scan or a pull scan. In the case of a pull scan, since the above-described exchange of authentication information is not required, the process proceeds to step S**805** as the scanning can be started. In the case of a push scan, the process proceeds to step S**803** in order to avoid leakage of the authentication information.
(45) In step S**803**, the CPU **601** determines whether or not the communication path between the processing terminal **101** and the image processing apparatus **102** is a P2P connection using a wireless LAN such as WiFi Direct. If it is a P2P connection, it is assumed that scanning can be started, and the process proceeds to step S**805**. On the other hand, if it is not a P2P connection, such as an environment for communicating on a typical LAN connection, the process proceeds to step S**804**.
(46) In step S**804**, the processing terminal **101** determines whether or not communication with the image processing apparatus **102** is encrypted. If the communication is not encrypted, such as in communication with an image processing apparatus **102** that supports only HTTP communication on port **80**, the process proceeds to step S**806**. If the communication is encrypted, the process proceeds to step S**805**.
(47) In step S**805**, the processing terminal **101** determines that communication with the image processing apparatus **102** is encrypted, transmits a request for instructing the image processing apparatus **102** to start the push scan as illustrated in FIG. **3**A, and ends the processing. On the other hand, in step S**806**, the processing terminal **101** determines that the communication with the image processing apparatus **102** is not encrypted, presents a warning screen as illustrated on the screen **707** to the user, and terminates the processing. The processing terminal **101** may periodically check the job status of the scan and present a completion display such as a screen **705** or a screen **706** to the user when the storage of the scan data is completed after step S**805**. The screen **705** is a completion display for when the cloud **105** is an external device separate from the processing terminal **101**, and the screen **706** is a completion display for when the destination of the scanning process is the processing terminal **101**.
(48) By such processing, it is possible to prevent information leakage by controlling whether or not to transmit a push scan start request packet including authentication information according to whether or not any of the communication mode for communication between the processing terminal and the image processing apparatus which is a scanner, the connection state between the processing terminal and the image processing apparatus, and the authorization level for communication between the processing terminal and the image processing apparatus satisfies a predetermined condition. In particular, when it is determined that the communication path between the processing terminal and the image processing apparatus is encrypted, it is possible to prevent leakage of authentication information by controlling not to transmit the scan start request packet.
(49) Configuration may be such that the processing terminal **101** does not restrict the start of the push scan when the user agrees to transmit the authentication information to the service **105**. That is, the process of starting the push scan may be continued according to the user's authorization that the authentication information may be transmitted in plaintext, such as when the processing terminal **101** and the image processing apparatus **102** are connected in a completely closed LAN. To this end, the processing terminal **101** can present a screen for confirming whether or not to continue the push scan process (for example, in the case where communication with the image processing apparatus **102** is not encrypted) to the user and acquire the selection. This processing is, for example, step S**906** (FIG. **7**B) of FIG. **9**, which will be described later, and when it is approved to continue the processing of the scan, it is assumed that the authorization level of the communication satisfies the above-described predetermined condition, and the start of the push

scan is not restricted. If it is not approved to continue the scan processing, it is determined that the authorization level of the communication does not satisfy the above-described predetermined condition, and the start of the push scan is restricted.
(50) FIG. **9** is a flowchart illustrating an exemplary process performed by the CPU **601** of the processing terminal **101** according to the present embodiment to display a screen for confirming whether to continue the push scan start process instead of the warning screen of FIG. **8**, and to restrict the push scan start. In the processing illustrated in FIG. **9**, the same processing as that illustrated in FIG. **8** is performed except that the processing of step S**901** and step S**902** is performed instead of step S**806**, and therefore, duplicated descriptions are omitted.
(51) In step S**901** performed when it is determined that the communication is not encrypted in step S**804**, the processing terminal **101** presents a confirmation screen to the user as to whether or not to continue the process for starting the push scan. In this example, a confirmation screen as illustrated in the screen **708** of FIG. **7**B is displayed, and the user's selection of whether to continue or cancel the processing is acquired. In step S**902**, the processing terminal **101** determines whether the user's selection of the confirmation screen displayed in step S**901** is to approve or cancel the continuation of the processing. When the continuation of the process is selected, the process transitions to step S**805**, and a push scan start instruction is performed. If cancellation of the process is selected, the process ends.
(52) When the start of the push scan is limited, the processing terminal **101** may instead suggest the user to perform the pull scan. In the case of a pull scan, as described above, it is not necessary to exchange authentication information, so the risk of leakage of authentication information can be avoided even in the case of plaintext communication. In this example, the processing terminal **101** presents to the user a screen for selecting whether or not to perform a pull scan instead of displaying a warning screen at step S**806**.
Second Embodiment
(53) The processing terminal **101**, which is an information processing apparatus according to the second embodiment, performs control so as not to perform a push scan by the image processing apparatus by restricting the display of an image processing apparatus whose communication is not encrypted in the list of search results in the search processing of the image processing apparatus performed in step S**401** to step S**405** according to the first embodiment. Except for this series of processes, the processing terminal **101** according to the present embodiment performs basically the same processing as that of the first embodiment, and therefore, a duplicated description thereof is omitted.
(54) The processing terminal **101** according to the present embodiment restricts the display of an image processing apparatus whose communication is not encrypted by not displaying the image processing apparatus in the list of search results, or by confirming whether the image processing apparatus is actually to be registered when the image processing apparatus is selected from the list. FIG. **11**, which will be described later, illustrates a flowchart for the case where a search result is not displayed in the list, and FIG. **12** illustrates a flowchart for the case where it is confirmed whether or not the registration is actually performed.
(55) The processing terminal **101** according to the present embodiment searches for image processing apparatuses by transmitting the search request in a multicast packet in the same manner as in step S**402**, for example. Here, the processing terminal **101** refers to mDNS response of the image processing apparatus to the search request and determines whether or not communication with each of the retrieved image processing apparatus is encrypted. This is determined, for example, by referring to the port number from the responses as illustrated in FIGS. **2**B to **2**D. In the following example, the processing terminal determines whether or not push scanning is possible based on whether or not communication is encrypted, but the determination may use a condition based on the connection form or a condition based on the authorization level.
(56) The processing terminal **101** can perform control so as not to display the image processing

apparatus that is determined not to encrypt the communication at the time of retrieval in the list of search results. The processing terminal **101** displays all the search results in a list, and when the image processing apparatus to be registered is selected by the user, if communication with the image processing apparatus is not encrypted, it may present something to that effect and acquire a selection of whether or not to continue the registration (for example, a screen **1005**). In this case, when communication with the selected image processing apparatus is encrypted, the selected image processing apparatus is registered in the list as a registered apparatus.

(57) FIGS. **10** to **12** are diagrams for explaining an example of processing performed by the processing terminal **101** according to the present embodiment at the time of searching for an image processing apparatus. FIG. **10** illustrates an example of a screen flow of the operation unit **605** displayed at the time of image processing apparatus search processing by the processing terminal **101** according to the present embodiment. FIG. **11** is a flowchart illustrating an exemplary process of restricting display when searching for an image processing apparatus, which is performed by the CPU **601** of the processing terminal **101** in the process as illustrated in FIG. **10**.

(58) Screen **1001** is an example of a screen for displaying a list of image processing apparatuses registered in the processing terminal **101**. The operation unit **605** can search for and register a new image processing apparatus by acquiring an operation of the user on the screen **1001**. In step S**1101**, the processing terminal **101** transmits the search request in a multicast packet. Here, when the user selects the “search” button on the screen **1001**, the CPU **601** transmits an mDNS search packet as illustrated in FIG. **2**A to the image processing apparatus, and the operation unit **605** performs a display indicating that the search is in progress as illustrated in the screen **1002**.

(59) In step S**1102**, the processing terminal **101** receives an mDNS response from each of the image processing apparatus that has performed the search. In step S**1103**, the processing terminal **101** determines whether or not the service of scanning the response packet uses encrypted communication. For an image processing apparatus that uses encrypted communication, step S**1104** processing is performed, and for an image processing apparatus that does not use encrypted communication, step S**1105** processing is performed. In step S**1104**, the processing terminal **101** adds an image processing apparatus using TLS encrypted HTTPS communication using the port **443** as illustrated in, for example, FIG. **2**B and FIG. **2**D to a list of search results as an apparatus in which communication is encrypted. On the other hand, in step S**1105**, the processing terminal **101** does not add an image processing apparatus that performs the HTTP communication scan service using the port **80** as illustrated in FIG. **2**C as an unencrypted apparatus to the list of search results. When the processing of step S**1104** or step S**1105** is completed, the processing proceeds to step S**1106**, and after the processing of step S**1103** to step S**1105** is repeated until mDNS search response is completed, the processing proceeds to step S**1107**. In step S**1107**, the processing terminal **101** displays a list of search results of the image processing apparatus on the display screen. The processing terminal **101** displays, for example, the image processing apparatuses **103** and **104** excluding the image processing apparatus **102** supporting only plaintext communication in the list.

(60) By such processing, it is possible to determine whether or not the path of communication with the processing terminal is encrypted at the time of retrieval of the image processing apparatus for registration. Therefore, by removing image processing apparatuses whose communication is not encrypted from the search result, it is possible to reduce the risk of leakage of authentication information.

(61) On the other hand, FIG. **12** is a flowchart illustrating an exemplary process performed by the CPU **601** of the processing terminal **101** in the process as illustrated in FIG. **10**, in which restrictions are performed after registering in the list of the searched image processing apparatuses. In step S**1201**, the processing terminal **101** transmits the search request in a multicast packet similarly to in step S**1101**. In step S**1202**, similarly to in step S**1102**, the processing terminal **101** receives an mDNS response from each of the image processing apparatus for which a search has

been performed.
(62) In step S**1203**, the processing terminal **101** adds each image processing apparatus that has returned an mDNS response to the list of search results. In step S**1204**, the processing terminal **101** determines whether or not the processing of step S**1203** has been performed on all the image processing apparatuses that have returned mDNS responses. If the processing has been performed for all image processing apparatuses, the process proceeds to step S**1205**; otherwise, the process returns to step S**1203**. The processing terminal **101** generates and displays a list of the search results of the image processing apparatuses as illustrated in the screen **1003** in step S**1205** after the reception of all the responses has been completed.
(63) In step S**1206**, the processing terminal **101** acquires the selection by the user of the image processing apparatus to be registered from the list. In step S**1207**, the processing terminal **101** determines whether or not the communication between the image processing apparatus selected in step S**1206** and the processing terminal **101** is encrypted in the same manner as in step S**1103**. If the communication with the selected image processing apparatus includes an encrypted scan service, the process proceeds to step S**1208**, and the processing terminal **101** adds the image processing apparatus to the registered list, and the process ends. On the other hand, if the selected image processing apparatus does not include an encrypted scan service, the process proceeds to step S**1209**.
(64) In step S**1209**, the processing terminal **101** displays a screen for acquiring a user selection as to whether or not to continue the registration process even though the encrypted communication is not included, as illustrated in the screen **1005**. In step S**1210**, the processing terminal **101** determines whether or not the user has selected to continue the registration process in step S**1209**. If continue is selected, the process proceeds to step S**1208**, and the processing terminal **101** adds the image processing apparatus to the registered list, and displays that the addition has been completed such as the screen **1006** to terminate the processing. If continuation is not selected, the registration process is canceled and the process ends.
(65) By such processing, it is possible to determine whether or not the path of communication with the processing terminal is encrypted at the time of retrieval of the image processing apparatus for registration. Then, the risk of leakage of the authentication information can be reduced by checking whether or not an image processing apparatus whose communication is not encrypted is actually to be registered when the image processing apparatus is selected from the list.
Third Embodiment
(66) In the printing system **100** according to the embodiment, push scanning in which the service **105** is used as a destination for a scan by the image processing apparatus **102** is performed. Meanwhile, when the image processing apparatus **102** performs a pull scan in which the destination of the scan data is the processing terminal **101**, it is not necessary to include the authentication information in the scan start instruction. From this point of view, the processing terminal **101** which is the information processing apparatus according to the third embodiment determines whether or not the image processing apparatus **102** can perform a pull scan. Next, the processing terminal **101** issues an instruction to start a pull scan when the image processing apparatus **102** is capable of pull scanning, and issues an instruction to start push scanning when a pull scan is not possible.
(67) The processing terminal **101** according to the present embodiment has the same configuration as that of the processing terminal **101** of the first embodiment except that when the image processing apparatus **102** is capable of a pull scan, the processing terminal **101** transmits a pull scan start instruction, and performs the same processing. Therefore, duplicate descriptions are omitted. In the present embodiment, the pull scan is a scan performed by the image processing apparatus **102** in response to a scan start instruction from the processing terminal **101**, in which the storage destination of the scan data is the processing terminal **101**.
(68) FIG. **13** is a flowchart illustrating an example of transmission processing of a pull scan start request performed by the processing terminal **101** according to the present embodiment. The

processing illustrated in FIG. **13** is started when an instruction to start scanning has been issued to the image processing apparatus **102** such as when the user presses a "Scan" button on the screen **703** of FIG. **7**, for example.
(69) In step S**1301**, the processing terminal **101** detects that the user has instructed the image processing apparatus **102** to start scanning. In step S**1302**, the processing terminal **101** determines whether the destination (storage destination) of the data of the scan by the image processing apparatus **102** is the self terminal, that is, the processing terminal **101**, or an external terminal such as the service **105**. Here, the processing terminal **101** makes the determination described above by referring to the item "Destination" inputted by the user on the screen **704** of FIG. **7**. If the destination is the self terminal, the process proceeds to step S**1303**, otherwise the process proceeds to step S**803** of FIG. **8** to perform subsequent processing on the push scan.
(70) In step S**1303**, the processing terminal **101** determines whether or not the processing terminal **101** is capable of performing pull scanning. Here, it is determined whether or not the processing terminal **101** is equipped with a pull scan function, and whether or not the enabled/disabled setting is enabled when the pull scan function is provided. If the processing terminal **101** is capable of pull scanning as well, the process proceeds to step S**1304**, otherwise the process proceeds to step S**803** of FIG. **8** to perform subsequent processing on the push scan. In step S**1304**, the processing terminal **101** transmits a start instruction to the image processing apparatus **102** to start scanning as a pull scan, and ends the processing.
(71) According to such processing, when pull scanning is possible in the present system, pull scanning can be started. In a pull scan, data can be transmitted and received by plaintext HTTP communication because the scan start request does not include authentication information. Therefore, leakage of the authentication information can be prevented.
Fourth Embodiment
(72) As illustrated in FIG. **2**A, the processing terminal **101** according to the first embodiment searches for a terminal having a scan service without distinguishing between push scanning and pull scanning. On the other hand, the processing terminal according to the present embodiment performs search processing assuming that "pull scan service" and "push scan service" exist in addition to the conventional "scan service".
(73) FIGS. **17**A to **17**C are examples of packets for mDNS service searching and response when "pull scan service" and "push scan service" exist in addition to "scan service". In comparison with the example of FIG. **2**, pull scan and push scan services are added to a packet from the processing terminal and a response packet from the image processing apparatus, and an image processing apparatus that only supports a conventional scan service may return the same response as in the example of FIG. **2**. FIG. **17**B is an example of a packet response from an image processing apparatus that performs plaintext communication (port number **80**) corresponding to a pull scan service. Also, FIG. **17**C is an example of a packet response from an image processing apparatus that performs encrypted communication (port number **443**) corresponding to both a pull scan service and a push scan service.
(74) In the present embodiment, a search request is made in each of a pull scan and a push scan, and a response is made to each of them. Therefore, even in an image processing apparatus in which "Use TLS" is disabled and "Push" is enabled as described in the below-described fifth embodiment, for example, if pull scanning is possible, a response with information indicating something to that effect can be returned.
(75) In the present system, the image processing apparatus transmits the search request from the processing terminal **101** including information as to whether or not a push scan is possible in the response. However, in consideration of prevention of leakage of authentication information, it is not necessary to indicate to the processing terminal **101** that a push scan can be performed when performing plaintext communication. For this reason, when communication with the processing terminal **101** is plaintext communication, the image processing apparatus according to the present

embodiment does not transmit to the processing terminal information that a push scan is possible in response to a search request from the processing terminal **101**. That is, information indicating that push scanning is not possible is transmitted as a response.

(76) By such processing, the image processing apparatus side is configured to perform a service response by each of a push scan and a pull scan, so the appropriate service response using a pull scan can be performed.

Fifth Embodiment

(77) An image processing apparatus **1600** according to the fourth embodiment performs push scanning in response to a push scan start instruction from the processing terminal **101**, similarly to the image processing apparatus **102** of the first embodiment. In addition, the image processing apparatus **1600** sets (enables/disables) whether or not to perform encrypted communication and sets (enables/disables) whether or not to perform a push scan according to the setting. Here, the image processing apparatus **1600** is set so as not to perform a push scan when encrypted communication is set to be disabled. That is, by linking the setting of encrypted communication with the setting of availability of push scan and not performing push scan when encrypted communication is not performed, leakage of authentication information is prevented.

(78) The image processing apparatus **1600** according to the present embodiment is implemented in a standard in which a “pull scan service” and a “push scan service” exist in addition to the “scan service” according to the fourth embodiment. Accordingly, the image processing apparatus **1600** returns a response based on the setting (enabled/disabled) of whether or not to perform the above-described encrypted communication and the setting (enabled/disabled) of whether or not to perform a push scan determined according to the setting in response to the search request from the processing terminal **101**. However, the image processing apparatus **1600** is not particularly limited to the implementation in this standard, and a response generated in the conventional standard as illustrated in FIG. **2** of the first embodiment may be performed.

(79) FIG. **16** is a block diagram illustrating an example of a hardware configuration of an image processing apparatus **1600** according to the present embodiment. The image processing apparatus **1600** includes a CPU **1601**, a ROM **1602**, a RAM **1603**, a storage unit **1604**, a printer processing unit **1605**, a scanner processing unit **1606**, a communication unit **1607**, and an operation unit **1608**. The CPU **1601** directly or indirectly controls each device (such as the ROM and the RAM) connected by an internal device and executes a program for implementing the invention. The ROM **1602** is a read-only storage device that stores programs executed by the CPU **1601** and stores BIOS as firmware. The RAM **1603** functions as the main memory or work memory of the CPU **1601** and is utilized to load software modules for implementing the invention. The storage unit **1604** is a storage area and is, for example, a hard disk drive (HDD) or a solid state drive (SSD) in which an OS or a software module that is basic software is stored. The scanner processing unit **1606** performs a process of reading and digitizing image files scanned by pressure plates or feeders. The printer processing unit **1605** controls copying or discharges an image printed on a designated image file in response to a print instruction from an external terminal. The operation unit **1608** functions as a display unit for displaying information to the user and a reception unit for receiving an instruction from the user. The operation unit **1608** is, for example, a liquid crystal display unit having a touch panel function or a display having various hard keys.

(80) The CPU **1601** controls the display of information and the reception of user operations in cooperation with the operation unit **1608**. The communication unit **1607** is an interface for the image processing apparatus **1600** to connect to the network. The communication unit **1607** according to the present embodiment is assumed to be a communication interface that performs wired communication based on Ethernet (registered trademark), but is not particularly limited thereto as long as communication is enabled. The communication unit **1607** can perform communication in the same manner as the communication unit **606** of the first embodiment.

(81) FIG. **14** is an example of a scan setting screen displayed on the operation unit **1608** of the

image processing apparatus **1600**. The user performs desired scan settings via the screen **1400** illustrated in FIG. **14**. In the screen **1400**, an item of "use network scan" which is a setting for whether or not to use the network scan function is set. The transmission and reception of data over the network by application functions such as IPP scanning or eSCL scanning is enabled/disabled according to the enabled/disabled setting for network scanning use. Further, when the setting of "use TLS" in the screen **1400** is set to enabled/disabled, transmission and reception of data via plaintext communication using the port **80** of HTTP is set to enabled/disabled. That is, when "Use TLS" is set to be enabled, data is transmitted and received by plaintext communication. Further, in the screen **1400**, as the type of the transmission type, "Pull" and "Push" are separately set to be enabled/disabled. Here, the pull scan function of the image processing apparatus **1600** is enabled when "Pull" is set to be enabled, and the push scan function of the image processing apparatus **1600** is enabled when "Push" is set to be enabled. Either one of the setting of "Pull" or the setting of "Push" may be set to enabled, or both may be set to enabled. After the user selects the various scan settings, the CPU **1601** acquires the selected various settings and stores them in the storage unit **1604** by pressing the "SAVE" button.
(82) FIG. **15** is a flowchart illustrating an example of a process performed by the image processing apparatus **1600** to prohibit the push scan setting when the encrypted communication is disabled. The description will be given below as if each process proceeds by user selection of items on the screen **1400**, but there is no particular limitation to this as long as the same setting is made. The process illustrated in FIG. **15** starts from step S**1501** upon acceptance of an instruction to save the scan settings such as a pressing of the "SAVE" button of the screen **1400**. In step S**1501**, the CPU **1601** detects the pressing of the "SAVE" button by the user, and performs subsequent processing using the settings acquired and stored upon that pressing as a processing target. In step S**1502**, the CPU **1601** determines whether the Use Network Scan option is enabled or disabled. If the Use Network Scan option is enabled, the process proceeds to step S**1503**; if disabled, the process proceeds to step S**1506**.
(83) In step S**1503**, the CPU **1601** determines whether the "Push" setting is enabled or disabled. If the "Push" setting is enabled, the process proceeds to step S**1504**; otherwise, the process proceeds to step S**1506**. In step S**1504**, the CPU **1601** determines whether the "Use TLS" setting is enabled or disabled. If the "Use TLS" setting is disabled, the process proceeds to step S**1505**; otherwise, the process proceeds to step S**1506**.
(84) If the "Push" setting is enabled and "Use TLS" is enabled, the authentication information included in push scan communication is communicated in plaintext. Therefore, in step S**1505**, the CPU **1601** controls the system so that push scan is not performed by the scan setting acquired in step S**1501**. Here, a warning message indicating that the combination of this setting is impossible is displayed on the display unit, and the processing is terminated without saving the acquired setting. If the "Push" setting is enabled and the "Use TLS" setting is not enabled, the CPU **1601** stores the acquired setting as something that is not a security problem in step S**1606**.
(85) In other words, although the image processing apparatus **1600** according to the present embodiment excludes performing both a push scan and plaintext communication, implementation is not limited thereto. For example, the image processing apparatus **1600** may display a warning message when the "SAVE" button is pressed as in the flowchart of FIG. **15**, or may enable the setting of "Use TLS" in conjunction with the setting of "Push" being enabled. Further, the image processing apparatus **1600** may automatically disable the "Push" setting when "Use TLS" is set to disabled, and may be grayed out so as not to be selectable, for example.
(86) According to this processing, when the push scan setting of the image processing apparatus is enabled, encrypted communication can be always performed. Therefore, when the push scan setting in the scan apparatus is enabled, linkage and prohibition setting processing are performed so that the encrypted communication setting is always enabled, such that the communication path including the authentication information is always encrypted, so that the authentication information

can be prevented from being eavesdropped.
(87) In the present embodiment, the setting as illustrated in FIG. **14** is performed in the image processing apparatus **1600**. However, the setting may be input on another apparatus; for example the setting may be inputted on an external apparatus such as the processing terminal **101**, and the image processing apparatus **1600** may perform the processing from step S**1502** for the scan setting when the "SAVE" button is pressed.
Other Embodiments
(88) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.
(89) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(90) This application claims the benefit of Japanese Patent Application No. 2021-036675, filed Mar. 8, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. A scanner that supports a setting for TLS (Transport Layer Security) of a communication via a network, the scanner comprising: at least one memory storing instructions, and at least one processor executing the instructions causing the scanner to: receive a request according to a predetermined protocol from an information processing apparatus, wherein the information processing apparatus executes a program that uses the predetermined protocol; and transmit, in response to the request, a response, wherein the response indicating that a predetermined function is possible is transmitted to the information processing apparatus, in a case where the setting for TLS of the scanner is enabled, and wherein the response not indicating that the predetermined function is possible is transmitted to the information processing apparatus, in a case where the setting for TLS of the scanner is disabled, and wherein the predetermined function is a function that requires the transmission of credential data from the information processing apparatus to the scanner via the network.
2. The scanner according to claim 1, wherein the credential data is a password.
3. The scanner according to claim 1, wherein the scanner receives credential data for the predetermined function from the information processing apparatus after the response indicating that the predetermined function is possible has been transmitted, and wherein the predetermined

function is a scan function and the credential data is used on the scanner at time of execution of the scan function.

4. The scanner according to claim 1, wherein the predetermined protocol is eSCL protocol.

5. The scanner according to claim 1, wherein the instructions further cause the scanner to display a setting screen that receives a user operation to enable or disable the setting for TLS.

6. The scanner according to claim 1, wherein the instructions further cause the scanner to transmit, in response to the request, a response indicating that the predetermined function is not possible to the information processing apparatus, in a case where the setting for TLS of the scanner is disabled.

7. The scanner according to claim 1, further comprising a printing device.

8. A method for a scanner that supports a setting for TLS (Transport Layer Security) of a communication via a network, the method comprising, receiving a request according to a predetermined protocol from an information processing apparatus, wherein the information processing apparatus executes a program that uses the predetermined protocol; and transmitting, in response to the request, a response, wherein the response indicating that a predetermined function is possible is transmitted to the information processing apparatus, in a case where the setting for TLS of the scanner is enabled, and wherein the response not indicating that the predetermined function is possible is transmitted to the information processing apparatus, in a case where the setting for TLS of the scanner is disabled, wherein the predetermined function is a function that requires transmission of credential data from the information processing apparatus to the scanner via the network.

9. The method according to claim 8, wherein the credential data is a password.

10. The method according to claim 8, wherein the scanner receives credential data for the predetermined function from the information processing apparatus after the response indicating that the predetermined function is possible has been transmitted, and wherein the predetermined function is a scan function and the credential data is used on the scanner at time of execution of the scan function.

11. The method according to claim 8, wherein the predetermined protocol is eSCL protocol.

12. The method according to claim 8, further comprising displaying a setting screen that receives a user operation to enable or disable the setting for TLS.

13. The method according to claim 8, further comprising transmitting, in response to the request, a response indicating that the predetermined function is not possible to the information processing apparatus, in a case where the setting for TLS of the scanner is disabled.

14. The method according to claim 8, wherein the scanner further comprises a printing device.

15. An image processing apparatus that supports a setting for TLS (Transport Layer Security) of a communication via a network, the image processing apparatus comprising: a printing device; a scanner; at least one memory storing instructions, and at least one processor executing the instructions causing the image processing apparatus to: receive a request according to a predetermined protocol from an information processing apparatus, wherein the information processing apparatus executes a program that uses the predetermined protocol; and transmit, in response to the request, a response, wherein the response indicating that a predetermined function is possible is transmitted to the information processing apparatus, in a case where the setting for TLS of the image processing apparatus is enabled, and wherein the response not indicating that the predetermined function is possible is transmitted to the information processing apparatus, in a case where the setting for TLS of the image processing apparatus is disabled, and wherein the predetermined function is a function that requires transmission of credential data from the information processing apparatus to the image processing apparatus via the network.